

Extending the Argument-Based Approach to Quantitative Studies

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💡 Chapter box

Central question: How can a reader extend the argument-based approach from a simple study to a relatively complex quantitative study with coordinated questions whose estimands, evidence, models, and interpretations interact?

Focal components: Every question–answer pair retains its own argument with linked statistical and substantive layers. To support that reading task, the chapter distinguishes multiple question-level arguments from multiple statistical strands inside one argument. It then extends the basic six-stage construction workflow to seven stages for coordinated questions, develops seven quality-evaluation questions, and shows how regression can instantiate question-specific estimands.

Main evidence: An upgraded hypothetical special-practice study with 320 randomized participants, three distinct questions, imperfect attendance, missing delayed outcomes, section structure, adjustment, and interaction; and a compact transfer application to the Deslauriers et al. published study and its supporting materials.

Scope: Reading quantitative studies remains the approach's **primary use**. Quality evaluation is its **secondary use**: it follows and depends on reading but is not part of reading's definition. The construction workflow is its **auxiliary use** for researchers conducting a study. *How to Read a Quantitative Study as an Argument* is sufficient for a simple study with one primary question, or with multiple independent questions that can be addressed separately. This advanced tutorial is optional and is not a prerequisite for Chapters 2–14. It addresses relatively complex studies with coordinated questions. Its construction workflow and quality-evaluation questions retain those ranked uses. The workflow covers much, but not all, of the research process; the evaluation questions do not produce a mechanical score; and regression does not supply measurement meaning, design support, substantive importance, or generalizability by itself. All learner-directed activities follow Section 9 in the unnumbered Exercises section.

1 Coordinating Question-Specific Arguments in More Complex Studies

1.1 What remains unchanged

The core rule remains:

Each question–answer pair has one argument with linked layers.

The basic tutorial can therefore be applied more than once when a study has several independent questions that can be read separately. The advanced treatment becomes useful when questions are coordinated through shared estimands, evidence, models, priorities, or synthesis, so that decisions and interpretations for one pair can affect another.

Within each statistical or substantive layer, Evidence and Justification support Inference, which supports a bounded Claim:

$$E + J \longrightarrow I \longrightarrow C.$$

The statistical layer addresses what the recorded evidence supports about a defined estimand. The substantive layer situates, augments and contextualizes, interprets, and bounds that result as an answer about the people, practices, settings, or phenomena named in the question.

Complexity does not create permission to merge questions that are owed separately reportable answers. If a study separately asks about an average assignment effect, variation in that effect, and the relationship between attendance and performance, it owes three separately reportable question-facing answers. The three question-answer pairs therefore have three arguments. They may share the same participants, design records, outcome, and fitted model, but shared support does not make their obligations identical.

1.2 What becomes more demanding

Four extensions matter in this chapter.

First, **several questions require coordination**. A primary question may be prespecified, a secondary question may ask about effect variation, and an exploratory question may arise after evidence production has begun. Such a new question starts its own explicitly exploratory argument cycle; it is not added retrospectively to a planned cycle. Priorities and analytic status must remain visible. Study-level synthesis should integrate question-specific answers without turning them into a new unsupported conclusion.

Second, **the evidence-generating process becomes less tidy**. Participants may enter through nonprobability recruitment, attend different amounts, miss an outcome, or be nested in sections. These features affect what counts as evidence, which estimand can be estimated, and how uncertainty should be calculated.

Third, **the three uses remain ranked and distinct**. Reading the arguments is the primary use. Quality evaluation is the secondary use because it judges what reading first identifies. The construction workflow is the auxiliary use for researchers building the support. Stages and evaluation questions are not additional argument layers.

Fourth, **regression can express several estimands in one mathematical language**. A group indicator can represent a randomized comparison; baseline variables can improve precision; an interaction can express effect variation. The equation does not supply design, measurement meaning, or substantive importance by itself.

1.3 Shared material, question-specific obligations

Suppose two questions use the same delayed score. The score records are shared STAT-E. Yet the average-effect question and the effect-variation question use those records differently. The latter additionally requires adequate variation and overlap in baseline preparation, an interaction specification, and enough precision to distinguish meaningful patterns.

Likewise, a common randomization record supports the assignment-effect question directly. It does not support a causal effect of attendance, because attendance occurs after assignment and is not itself randomized. The role and adequacy of support are always question-specific.

The same discipline applies to conclusions. “Assignment improved performance on average” does not imply “the improvement was larger for lower-preparation students.” An inconclusive interaction is not evidence that effects are equal. A positive average effect and uncertain variation can coexist without contradiction.

1.4 One question can contain several statistical strands

A different structure arises when several statistical estimands are jointly needed for one integrated substantive answer. Each estimand then has a **statistical strand** within the pair’s one statistical layer:

$$STAT-E_k + STAT-J_k \longrightarrow STAT-I_k \longrightarrow STAT-C_k.$$

The strands feed one substantive layer and one question-facing answer. An estimand counts as a strand when it is subordinate to one prespecified integrated answer obligation and is not itself owed or interpreted as a standalone substantive conclusion. Separately owed conclusions form separate pairs. Independent answerability is a useful diagnostic, but the number of hypotheses, coefficients, tests, estimands, tables, or pooled estimates does not decide. A hypothesis is a proposition about one or more estimands, not itself an estimand. Neither hypothesis count nor estimand count mechanically determines the number of pairs or strands. Author wording and analytic status are cues to the intended obligations, not permission to merge conclusions that the study actually owes separately. The display may compress shared support as clarity permits; it cannot compress separately owed conclusions into strands.

For a compact illustration, imagine that the upgraded study had asked one integrated decision question: **Did assignment meet the prespecified success criterion of a positive delayed-performance effect and no completion-rate reduction greater than five percentage points?** One strand addresses the score effect; the other addresses the completion-rate difference and its prespecified non-reduction margin. Nonsignificance or nearly equal observed rates alone would not establish non-reduction. The strands share an assignment-and-design core but retain estimand-specific evidence, justification, inferences, and statistical claims. SUB-I integrates the two results, and SUB-C says whether the single success criterion was met

while preserving both sets of conditions and limitations. The two statistical claims must still be reported transparently, but here they are inputs to that one decision rather than standalone substantive conclusions. If the study instead owed a separate question-facing conclusion for either outcome, it would have separate pairs. The formulation is a granularity illustration, not a fourth worked question below.

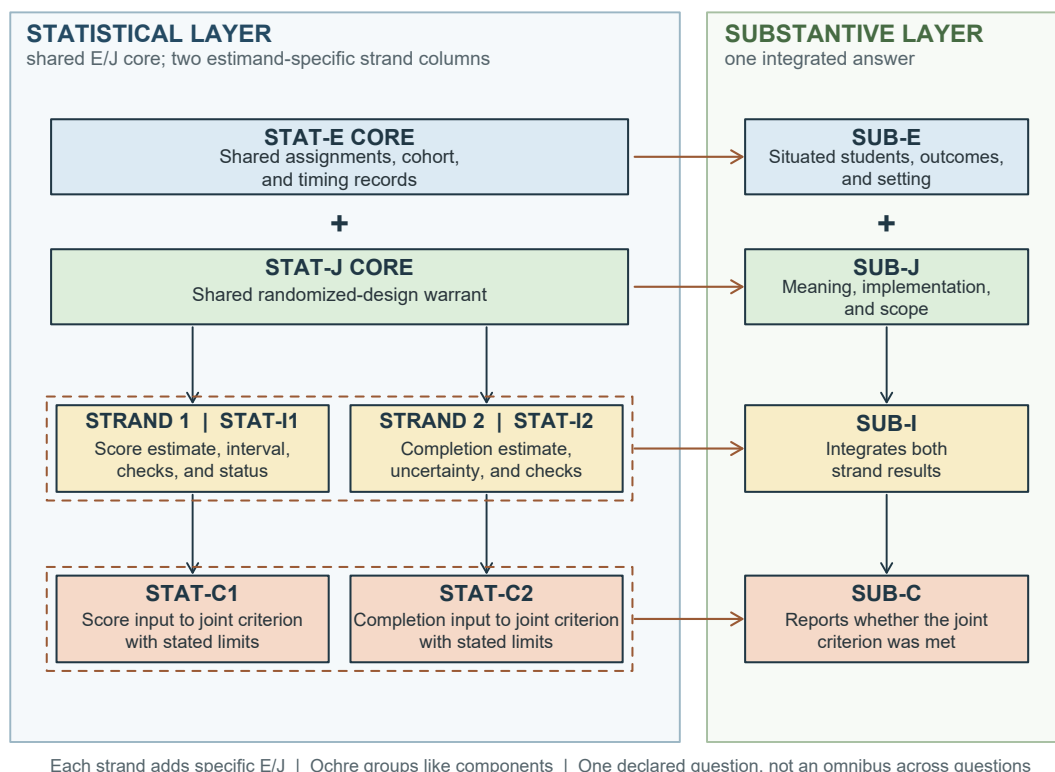


Figure 1: One question-level argument with two statistical strands and one substantive layer.

Parallel strands need not be statistically independent. Their E and J content may have a shared core and estimand-specific additions, as the figure shows; a common documentary trace can serve both without becoming either component. Several estimators, checks, or sensitivity analyses for the same estimand also do not create additional strands. Chapters 3 and 4 develop the formal estimand and comparison distinctions.

2 The Upgraded Special-Practice Study

2.1 The study record

The university now implements the study in four first-year course sections. Each section contains 80 consenting students. Within every section, 40 students are randomly assigned to Special Practice and 40 to Usual Practice, giving 320 randomized participants in total.

As in the basic tutorial, *Special Practice* is a neutral name for the hypothetical format rather than a conclusion about its quality.

Before assignment, students complete a 0–40 baseline-preparation assessment. Both formats contain four scheduled sessions matched in duration and topic coverage. Attendance is recorded initially as implementation information, not as the focus of a research question. A 0–40 delayed assessment is scheduled two weeks after the final session.

Unlike the basic example, not every participant attends most sessions or completes the delayed outcome.

Table 1

Descriptive record for the upgraded special-practice study.

Feature	Special Practice	Usual Practice
Assigned students	160	160
Baseline score, M (SD)	20.3 (5.6)	19.9 (5.8)
Attended at least three sessions, n (%)	118 (73.8%)	126 (78.8%)
Delayed outcome observed, n (%)	150 (93.8%)	150 (93.8%)
Observed delayed score, M (SD)	27.2 (5.4)	24.1 (4.6)

The two observed-outcome means differ by 3.1 points. That difference describes the 300 observed outcomes. It is not automatically the primary answer for all 320 randomized students because outcome missingness and the planned estimand must still be addressed.

2.2 Three question–answer pairs

The study has three explicitly separated questions.

Primary, prespecified assignment question

Among all 320 randomized participants, what was the average effect of assignment to Special rather than Usual Practice on the delayed score?

The estimand is the average assignment effect for all randomized participants. The analysis uses the prespecified multiple-imputation and standardization procedure described in Section 3.5. It adjusts for baseline and section while retaining all 320 participants in the population named by the estimand. A two-point difference was identified prospectively as a potentially important benchmark.

The estimated assignment effect is 2.9 points, 95% interval [1.9, 4.0]. The report gives a positive average assignment-effect answer under its stated conditions. Because the interval includes values just below and above two points, the reported answer does not claim that the effect exceeded the importance benchmark.

Secondary, prespecified effect-variation question

How did the Special-minus-Usual assignment effect vary with baseline preparation?

The fitted interaction estimates that the assignment effect changes by -0.8 points for each ten-point higher baseline score, 95% interval [-2.7, 1.1]. The compatible values include a meaningfully decreasing pattern, little or no variation, and an increasing pattern. This analysis uses the same imputed datasets and the same 320-person target cohort as the primary analysis. The reported answer therefore does not resolve the direction or substantive size of effect variation.

Exploratory attendance question

After evidence production had begun, the researchers formulated a new attendance question. It therefore began a separate, explicitly exploratory argument cycle rather than being inserted into the prospectively coordinated cycles for Pairs 1 and 2. The new cycle reused attendance, assignment, baseline, section, and outcome records, but its question and analytical plan were retrospective to those already produced records.

After accounting for assignment, baseline preparation, and section, how was the number of attended sessions associated with delayed score?

The adjusted association is 0.3 points per additional session, 95% interval [-0.4, 0.9] among the 300 students with observed delayed outcomes. Modest negative and positive observed-case associations both remain compatible with the result, so it does not resolve direction. More importantly, attendance was not randomized, and this exploratory analysis does not address outcome missingness beyond restricting the analysis to observed cases. Even a precise positive association would not automatically estimate the causal effect of attending one more session.

2.3 A question-to-argument map

Table 2

Three question–answer pairs and their distinct arguments.

Pair	Estimand and status	Result	Reported answer and stated bounds
1. Assignment effect	Average effect of assignment among all 320 participants; primary and prespecified	2.9 points, 95% interval [1.9, 4.0]	Positive average assignment effect under the missingness, implementation, and model conditions; exceeding two points remains uncertain
2. Effect variation	Change in assignment effect per ten-point higher baseline score among all 320 participants; secondary and prespecified	-0.8 points, 95% interval [-2.7, 1.1]	The result does not resolve the direction or substantive size of effect variation
3. Attendance association	Conditional association per additional attended session among 300 observed-outcome cases; exploratory	0.3 points, 95% interval [-0.4, 0.9]	Modest negative and positive observed-case associations remain compatible; the analysis does not establish a causal effect of attendance

The three arguments share recruitment, assignment, baseline, attendance, and outcome records. They do not share every justification. Random assignment supports Pair 1 and the assignment component of Pair 2. It does not randomize baseline preparation or attendance. Pair 2 needs interaction-specific support. Pair 3 needs an associational interpretation and explicit protection against causal wording.

They are three pairs because the study owes a separately reportable, question-facing conclusion for each one, not simply because the estimands or analytic statuses differ. Independent answerability helps diagnose that structure, while the authors’ labels help locate the obligations; neither is a mechanical counting rule.

2.4 The primary argument across both layers

The primary question can be mapped without repeating the full eight-component table from the basic tutorial.

STAT-E contains records for all 320 assignments, baseline scores, sections, attendance, outcome availability, delayed scores, implementation, and scoring. The 20 missing outcomes remain part of the evidence record even though their scores are unavailable.

STAT-J records the study’s offered reasons connecting the within-section randomization and all-randomized estimand to baseline and section adjustment, the prespecified multiple-imputation procedure, standardization, interval-combination method, and planned checks. Its force depends on implementation, absence of material interference, measurement comparability, and a plausible connection between observed information and missing outcomes.

STAT-I brings together the 2.9-point standardized estimate, 95% interval [1.9, 4.0], the completed missingness stress test reported in Section 3.5, model checks, confirmatory status, and the two-point benchmark. **STAT-C** states a positive average assignment contrast for the 320-person estimand under those conditions while retaining uncertainty about whether the contrast exceeds two points.

Across layers, **SUB-E** situates the records as students in four sections, assigned to two matched-time practice formats and assessed on a common delayed course measure. **SUB-J** augments and contextualizes the statistical justification with the intended educational meaning of the formats and assessment, implementation evidence, plausible alternatives, the importance benchmark, and limits on population transport.

SUB-I interprets the 2.9-point estimate and interval on the 0–40 delayed assessment scale. It considers whether that magnitude is important without declaring the benchmark crossed. **SUB-C** records the study’s participant- and implementation-bounded causal answer; Section 6 evaluates whether it is supported.

This component map also shows why the observed-case difference of 3.1 and the primary estimate of 2.9 are not competing answers. They describe different analytical objects. One is a raw comparison among observed outcomes; the other addresses the all-randomized estimand through the prespecified adjusted and standardized procedure.

2.5 Study-level synthesis

The report gives a study-level synthesis that integrates the three answers without erasing their differences:

Assignment to Special Practice improved delayed-assessment performance on average in this implementation. The study does not establish that the average improvement exceeded two points or that the assignment effect varied with baseline preparation. The observed-case attendance association was inconclusive, and the analysis does not establish that attending an additional session caused better performance.

The synthesis does not invent a fourth quantitative result. If the authors wished to claim that Special Practice works specifically by increasing attendance, that mechanism claim would require another question and argument.

3 Extending Evidence, Measurement, Estimands, and Missingness

3.1 Evidence is produced, not merely available

Evidence is not a neutral table handed to a statistical method. It records who or what was observed, how observations were produced, which comparison the design created, and which intended cases are absent.

For the upgraded study, STAT-E includes:

- eligibility, consent, section, and assignment records for all 320 students;
- baseline and delayed assessment scores and their scoring rules;
- session schedules, delivery, and attendance;
- the identities and timing of the 20 missing delayed outcomes; and
- processing and inclusion decisions.

The means and intervals are derived results in STAT-I, not the empirical evidence itself. This narrower use of *Evidence* keeps the inferential move visible even though everyday writing often calls results “evidence.”

3.2 Population, unit, case, and level

The **target population or target set** contains the entities, settings, or occasions to which an intended answer refers. The **sampling frame** is the reachable set from which participants can arise. The **analytic records** are the cases and observations used by a stated procedure. These sets need not coincide.

The direct target cohort for Pair 1 is the 320 consenting participants. The study does not use probability sampling from all first-year students. Random assignment can support a causal comparison for these participants while wider generalization remains uncertain.

The **unit** is the entity about which the question is asked. Here the student is the unit of assignment and outcome measurement. Section is a randomization block and a contextual grouping. Repeated records for one student remain observations on one unit; they are not independent students.

Level matters because a relationship among students does not automatically describe sections or institutions. Four sections also provide little direct evidence about variation across the larger

population of possible sections. Including section indicators respects the blocked comparison but does not justify a general claim about all course contexts.

3.3 Estimands keep questions distinct

Recall the Chapter 1 contrast: the **estimand** is what the study seeks to learn, whereas the **estimate** is the numerical result obtained from observed evidence. The advanced point is that an estimand can answer a question directly or contribute as one strand of an integrated answer. Defining estimands before choosing estimators helps connect substantive questions, design and assumptions, and empirical evidence (Lundberg et al., 2021).

Pair 1 has an estimand of the average assignment effect for all 320 participants. Pair 2 has an estimand describing variation in that assignment effect with baseline preparation. Pair 3 has an estimand of an observed conditional association with attendance. The same outcome and regression family neither make these one estimand nor decide the number of question–answer pairs.

Variable roles also come from questions. Baseline preparation is a precision variable in Pair 1, a moderator in Pair 2, and an adjustment variable in Pair 3. Attendance is an implementation record for Pair 1 and the focal explanatory variable in Pair 3. Its role cannot be inferred merely from its column position in a dataset.

Table 3

Question-specific evidence and estimand obligations.

Pair	Evidence and estimand obligation	Main vulnerability	Consequence for the claim
Assignment effect	Preserve all 320 randomized participants in the target set and use baseline, section, implementation, and outcome records appropriately	Twenty delayed outcomes are missing; participation is from one university setting	The answer remains conditional on missingness handling and limited to this implementation and represented participants
Effect variation	Retain the randomized contrast while estimating how it changes over the supported baseline range	Interaction form, limited precision, and sparse values near the baseline extremes	Do not claim a subgroup pattern from the negative point estimate

Pair	Evidence and estimand obligation	Main vulnerability	Consequence for the claim
Attendance association	Define attendance consistently and preserve its post-assignment timing	Motivation, difficulty, health, and other factors can affect both attendance and score	Report association only; do not inherit the causal support of assignment

3.4 Observable measures still require interpretation

Both assessments are directly recorded, but neither is theoretically transparent. Their meaning depends on content coverage, administration, scoring, reliability, and evidence supporting the intended interpretation (American Educational Research Association et al., 2014; Kane, 2013).

The delayed score is evidence about performance on a defined course assessment. It may support a bounded interpretation of delayed learning over the assessed content, but it does not measure every kind of knowledge, transfer, motivation, or long-term retention. Baseline preparation likewise represents performance on a particular instrument, not a complete amount of preparation possessed by each student.

Measurement comparability matters across assignment groups and baseline levels. If one format teaches test-specific cues unavailable in the other, the observed score difference may still be real while its interpretation as broader learning becomes weaker. If the baseline scale behaves differently near its ends, the interaction may partly reflect measurement rather than genuine effect variation.

3.5 Missing outcomes are part of the evidence-generating process

Ten students in each assigned group lack the delayed outcome. Equal counts do not make missingness harmless. The reasons could differ by baseline preparation, section, attendance, assignment, or an unobserved outcome.

The primary estimand remains the average assignment effect for all 320 randomized participants. An analysis of only the 300 observed outcomes estimates an observed-case contrast; it does not by itself change the declared estimand. Without conditions connecting observed cases to all randomized cases, however, that analysis does not identify the declared estimand.

The prespecified missing-outcome procedure uses assignment, section, baseline preparation, the assignment-by-baseline relationship, attendance, and available administrative information. It is designed to preserve the information needed for both the average-effect and effect-variation questions while keeping all 320 randomized participants represented in those estimands.

Attendance helps inform the missing outcomes; it neither defines the assignment estimand nor becomes a randomized exposure. Later chapters develop the imputation, model-fitting, standardization, and uncertainty-combination mechanics.

This procedure assumes that, after accounting for the recorded predictors, the unobserved delayed score does not add further information about whether it is observed. That condition cannot be proved from observed data. Sensitivity analysis should vary plausible departures and carry consequential changes into STAT-C and SUB-C.

One completed stress test subtracts four points from each imputed missing Special Practice outcome while leaving the Usual Practice imputations unchanged. The standardized estimate becomes 2.6 points, 95% interval [1.5, 3.7]. This deliberately unfavorable departure retains a positive direction while leaving the exact magnitude uncertain and whether the effect exceeds the two-point benchmark unresolved. The next question is a tipping-point or claim-change question: how different would plausible unobserved outcomes or assumptions need to be before the bounded answer changed direction, crossed the two-point benchmark, or otherwise changed materially? A departure that leaves materially important benefit and harm both compatible would require an inconclusive directional answer. That judgment depends on the question and meaningful benchmarks, not merely on whether an interval crosses zero. Sensitivity is not a competition to find a preferred result; it describes what the claim can withstand.

3.6 Documentary trace and status

An argument component names a **logical function**; study-specific content fills that function, and documentary traces show where the content can be inspected. The components are not filing categories for documents. A shared protocol can trace several functions without establishing that every reported action was performed correctly or that every offered reason is sound.

A reader uses Chapter 1's four **documentary statuses**—**stated in the paper**, **located in related materials**, **implicit but traceable**, and **not located**—for the most direct available trace of relevant component content.

A contradiction across sources is a separate flag, not a fifth status. These statuses describe what can be inspected; they do not evaluate its quality. A protocol can establish that an estimand was prospectively stated without proving that it was well chosen, and a complete script can establish computational traceability without establishing measurement quality. Conversely, **not located** does not prove that an underlying action never occurred.

Consider a deliberately mistaken alternative report, **not the upgraded study's actual account**: "Because practice-format assignment was randomized, we interpret the attendance association causally." Reading records that offered reason in **STAT-J**, with **stated in the paper** as its documentary status. It remains an identifiable justification because that is the function the authors give it, even though the reason is mistaken. Evaluation separately finds

that assignment to a format did not randomize attendance. Sharing the assignment record therefore does not supply the same justification for these different inferences.

Subject expertise helps readers recognize an offered function and make a conventional implicit connection explicit when it is already traceable in the record. Recognizing that connection is distinct from deciding whether it is correct or sufficient. Expertise cannot supply an undocumented study-specific premise, such as assuming that attendance itself was randomized.

A reader-added premise may be named during evaluation only as a proposed repair. It does not count as support in the argument currently being read. The word **proof** is therefore usually too strong: readers locate component content through documentary traces and then evaluate adequacy separately.

4 Research Design and the Claims It Can Support

4.1 Representation and comparison are different

Research design makes an observed comparison informative. It supplies distinguishable support for representation, comparability, timing, measurement, and implementation. A study can be strong in one dimension and weak in another (Shadish et al., 2002).

Random sampling primarily concerns who enters from a population and can support population generalization when selection probabilities and participation are handled appropriately. **Random assignment** concerns which condition participants receive and can support causal comparison within the study. One does not imply the other.

The upgraded study uses nonprobability participation and random assignment. Its design can support an assignment-effect answer for the 320 participants, but extension to other students, courses, institutions, or implementations needs additional evidence.

4.2 Design families and intended claims

An **observational study** observes conditions that arise without researcher assignment. It can directly support description, association, and prediction. Causal interpretation needs additional design-and-assumption support—often called **identification support** because it explains why the intended causal comparison can be learned from the observed comparison. Regression adjustment alone does not create it.

A **randomized experiment** uses chance to assign conditions. Randomization supports a causal comparison of assigned conditions when implementation, outcome observation, interference, and analysis are handled appropriately.

A **quasi-experiment** uses a threshold, timing change, policy rule, or comparison structure to approximate an experiment without random assignment. Its causal support comes from design-specific conditions rather than the word *quasi* or the presence of regression.

The strongest claim mode follows the question and design, not the complexity of the model. Description, association, prediction, and causal comparison can all use simple or elaborate analyses.

4.3 Blocking, dependence, and interference

Assignment in the upgraded study occurs separately within four sections, so section is a randomization block rather than the unit of assignment. The analysis and uncertainty procedure should respect the within-section comparison. A generic cluster correction based on only four sections is not an automatic substitute; later chapters develop the relevant methods.

Students may interact within sections. **Interference** occurs when one student's assigned format affects another student's outcome. Separate sessions and monitoring may make material interference less plausible, but peer discussion outside sessions could remain. The estimand should name the assignment strategy actually implemented; the claim should retain any unresolved spillover reservation.

4.4 Assignment is not attendance

Randomization most directly supports the effect of **assignment**. Attendance occurs after assignment and can be influenced by motivation, scheduling, difficulty, health, or early reactions to the format. Comparing students who attend different numbers of sessions generally loses randomization.

The attendance association is still useful descriptively. It can identify an implementation pattern, motivate a future attendance intervention, or support prediction. It cannot be promoted to the effect of receiving an additional session without stronger assumptions and design.

4.5 Analytic status and multiplicity

The primary and secondary questions were specified before outcomes were examined; the attendance question is exploratory. These statuses affect how the results should be calibrated but do not predetermine whether their answers are positive or inconclusive.

Several outcomes, subgroups, interactions, and model variants create multiple opportunities to find an interesting result. Multiplicity can be addressed through priority, prespecification, adjusted uncertainty where appropriate, transparent reporting, and independent confirmation. It cannot be solved by reporting only the model that produced the most favourable estimate.

4.6 Inferential conditions belong to particular arguments

Conditions should be linked to the inferential move they support rather than recited as one universal checklist.

Estimand and specification conditions ask whether the population, contrast, scale, functional form, and variable roles match the question. Pair 1 requires an all-randomized average assignment estimand; Pair 2 additionally requires a defensible scale and form for effect variation.

Evidence-production conditions concern recruitment, assignment, timing, measurement, implementation, and processing. Within-section randomization supports the assigned comparison, but only if the assignment record is preserved and the outcome has a comparable meaning across formats.

Uncertainty conditions identify the variation represented by the interval. The procedure should respect blocked assignment and student-level units. An interval can be numerically narrow while referring to the wrong estimand or ignoring a dependency created by design.

Selection and missingness conditions connect the observed records to the declared population. The primary argument needs an explicit reason why its missing-outcome procedure can address the all-randomized estimand and a sensitivity analysis for plausible departures.

Analytic-conduct conditions concern priority, prospectiveness, model changes, multiplicity, and reuse of the same evidence. A secondary prespecified interaction and an exploratory attendance analysis should not be reported as if they had the same evidential status as the primary contrast.

Checking and sensitivity conditions ask whether the main answer survives plausible alternatives. Diagnostics may reveal poor functional form or influential observations, but passing a small set of checks does not certify a model as true. The consequence of a material failure is a changed inference or claim, not merely an additional appendix figure.

Table 4

Design support and remaining claim limits in the upgraded study.

Design feature	Support supplied	Remaining limitation
Within-section random assignment	Average causal comparison of assigned formats for participants	Implementation, missing outcomes, interference, and transport
Nonprobability participation	Direct evidence about consenting participants	Wider population representation requires additional support
Baseline measurement	Precision adjustment and a moderator for a secondary question	It does not identify a causal effect of preparation

Design feature	Support supplied	Remaining limitation
Attendance recording	Implementation description and an exploratory association	Attendance is post-assignment and not randomized
Common delayed assessment	Comparable observed outcome under stated scoring	Score meaning remains bounded to the assessment and conditions
Prospective priorities	Distinguishes primary, secondary, and exploratory analyses	Prespecification does not guarantee adequate design or inference

Statistical analysis can express these design-supported estimands and improve precision. It cannot manufacture probability sampling, randomization of attendance, absence of interference, comparable measurement, or prospective status.

5 Extending the Construction Workflow

5.1 A seven-stage extension

The argument components describe what logically supports one answer. A **construction-workflow stage** describes an activity used to construct or revise that support. Components and stages therefore do different work.

The seven stages below extend Chapter 1’s six-stage workflow for studies with coordinated questions. The diagram compresses a coordinated set of question-level cycles; it does not create one omnibus argument cycle for the study. The extension covers much of the question-to-answer research process, but, as Chapter 1 notes, it is not a complete research-lifecycle model.

Basic Stages 1–3 remain advanced Stages 1–3, with prospective cross-question alignment added to Stage 3. Basic Stages 4–6 remain advanced Stages 4–6. New Stage 7 gives every question an explicit outcome and synthesizes across those outcomes without merging their arguments.

Table 5

How the basic six-stage workflow extends to seven stages for coordinated questions.

Basic six-stage workflow	Advanced seven-stage extension
1. Review the problem, setting, and relevant literature	1. Review literature and the problem for the coordinated question set
2. Frame the question and, where useful, a candidate answer	2. Frame questions and candidate answers, priorities, and analytic status

Basic six-stage workflow	Advanced seven-stage extension
3. Align design, measurement, estimand, evidence-production or acquisition plan, and inferential plan	3. Plan and align prospectively across questions: design, measurement, estimands, evidence production or acquisition, stopping, inference, checking, uncertainty, and sensitivity
4. Produce or acquire, and document, empirical and contextual evidence	4. Produce or acquire and document shared and question-specific empirical and contextual evidence
5. Estimate, check, and analyse	5. Estimate or predict, check, conduct sensitivity analysis, and reason statistically for every question-specific estimand
6. Contextualize, interpret, and complete the bounded answer	6. Contextualize, justify, interpret, and complete each bounded substantive answer
New for coordinated questions	7. Give every question an explicit outcome and synthesize across questions

The workflow adapts the iterative character of statistical enquiry emphasized by PPDAC and statistics-education guidance (American Statistical Association, 2016; Wild & Pfannkuch, 1999). Iteration is confined to the phase currently under construction. Phase 1 Stages 1–3 can iterate before a one-way handoff to Phase 2. Phase 2 Stages 4–7 can then iterate while evidence is produced, analysed, interpreted, and synthesized. Preregistration or another time-stamped plan can strengthen the record of prospective decisions, but it does not guarantee that the question, design, or analysis is adequate (Nosek et al., 2018).

Table 6

The seven-stage extension applied to the upgraded special-practice study.

Phase and stage	Main activity	Application to the upgraded study
Phase 1 – 1. Review literature and the problem	Develop prospective statistical and substantive justification	Examine relevant learning theory and prior evidence about attempted problem solving, feedback, delayed performance, assessment meaning, implementation, and likely moderators

Phase and stage	Main activity	Application to the upgraded study
Phase 1 – 2. Frame questions and candidate answers	Specify every question–answer obligation and its priority. Declare separate questions, priorities, and analytic statuses when conclusions will be reported separately; declare one question with jointly necessary estimands only when one integrated conclusion is owed.	In the initial coordinated cycles, declare the primary assignment question, secondary baseline-variation question, two-point benchmark, and candidate positive average effect
Phase 1 – 3. Plan and align prospectively across questions	Specify and align evidence production or acquisition, any stopping rule, estimands, design, measurement, inference, uncertainty, checking, missingness handling, sensitivity, and planned synthesis across pairs	Align the planned Pair 1 and Pair 2 cycles: randomize within section, stop recruitment after 320 participants are randomized without outcome-dependent stopping, measure baseline, define the delayed outcome, retain all participants in the population named by the primary estimand, and plan interaction and sensitivity analyses
Phase 2 – 4. Produce or acquire and document evidence	Produce STAT-E or obtain and designate question-specific analysis records; retain the contextual records needed for SUB-E	Record consent, assignment, delivery, attendance, baseline, outcomes, missingness reasons, and section context
Phase 2 – 5. Estimate, check, and analyse	Develop STAT-I and a provisional STAT-C for each pair	Estimate and check the assignment contrast and interaction; in the later exploratory attendance cycle, analyse the reused records only after separately framing its associational estimand
Phase 2 – 6. Contextualize, justify, interpret, and complete each bounded substantive answer	Complete the substantive reasoning and bounded answer for every pair	Complete the planned average-effect and effect-variation answers and the separate noncausal attendance answer with their distinct estimands, status, uncertainty, and limitations

Phase and stage	Main activity	Application to the upgraded study
Phase 2 – 7. Give explicit outcomes and synthesize	Record an explicit outcome for every question, then integrate those outcomes without erasing distinctions	Bring the completed cycles together: record a positive average assignment effect, inconclusive effect variation, and an inconclusive noncausal attendance association before writing the bounded study-level synthesis

For each question, Stages 1–6 constitute its own argument-construction cycle. The planned Pair 1 and Pair 2 cycles share activities and are aligned in Stage 3. Once Stage 6 has completed their bounded answers, Stage 7 records their outcomes and coordinates the study-level synthesis. The attendance question, which arose after Phase 2 had begun, first requires a separate exploratory cycle before its answer may enter that synthesis. Stage 7 does not create a fourth argument or complete the three arguments a second time.

The Phase 1 anchor for each cycle includes its question and mode, design, measurement, estimand, evidence-production or acquisition plan, stopping rule, and inferential plan. A prospectively documented candidate answer may be contradicted by the results without starting a new cycle. Stage 4 begins when evidence production starts or, for existing data, when the question-specific analysis records or extract are obtained or designated. First access to relevant outcomes or unblinding is a separate timing landmark: it may precede, coincide with, or follow that handoff. Prospectiveness is always a relation between a decision and the evidence relevant to that decision.

Figure 2 emphasizes iteration within each phase and a one-way handoff from Phase 1 to Phase 2.

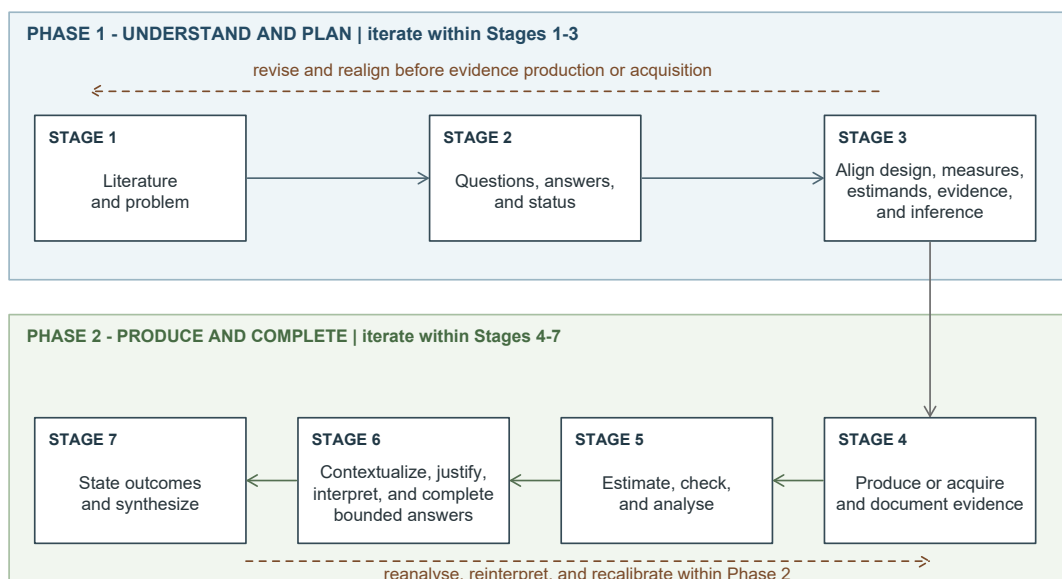


Figure 2: A compressed seven-stage view of coordinated question-level cycles, with iteration confined to each phase.

The dashed Stage 7-to-Stage 4 arrow means that synthesis may reveal a need to recheck evidence, analysis, or interpretation within Phase 2 of one or more affected question-level cycles. It does not send one merged study cycle back to Stage 4.

i Argument, cycle, and execution

An **argument** is the logical support for one answer obligation. A **workflow cycle** is the documented sequence used to construct that argument. An **execution or data attempt** is one operational production or acquisition of evidence within the cycle.

After an operational failure, evidence production or acquisition may be re-executed within the same cycle only when the Phase 1 anchor—including its acquisition plan and stopping rule—is unchanged, the reason is independent of the observed answer, and every attempt remains documented. Otherwise, the change requires an explicit outcome for the existing cycle and a new or revised cycle. Prospective status belongs to particular decisions relative to particular evidence; it is not inherited automatically.

5.2 Disciplined iteration

Iteration is disciplined when it stays within the current phase, responds to a documented problem, and changes analytic status or claims transparently.

Suppose a pilot reveals that reminder messages differ across formats and could affect attendance. Within Phase 1, before the focal study, the team can return from Stage 3 to Stages 1 and 2, clarify what belongs to each format, standardize reminders, and realign the coordinated question set within Stage 3.

Suppose instead that one section experiences a scheduling disruption after implementation begins. Phase 2 has started, so the team cannot erase the disruption or treat the old plan as if implementation had not occurred. It should document the disruption in STAT-E, examine a prespecified or clearly labelled sensitivity analysis in STAT-I, and narrow the claim if the result depends on that section. These actions iterate within Stages 4–7.

If Phase 2 reveals that a Phase 1 question, design, measurement, estimand, evidence-production or acquisition plan, stopping rule, or inferential plan must change, the same workflow cycle does not return to Phase 1. The team gives that cycle an explicit outcome and begins a new or revised argument cycle. Records, code, decisions, and other materials from the earlier cycle may be reused with their provenance and documentary status intact, but they do not automatically retain prospective status in the new cycle.

Applied to the upgraded study, the planned delayed-score pair keeps its own bounded answer. If the team comes to prefer a newly introduced outcome after evidence production, that outcome requires a separate exploratory pair and cycle rather than becoming a replacement primary result. If the planned delayed assessment cannot answer the original question, the current attempt is **not answerable as planned**; a redesigned study begins a new or revised cycle while reusing suitable earlier material transparently.

A within-phase return is not a licence to search until a desired answer appears. An apparently post hoc change can retain prospective status only when dated records show that it preceded the evidence relevant to that change. A question or plan chosen after inspecting existing relevant records is not prospective to those records; a plan fixed before genuinely new evidence may be prospective for that evidence while remaining explicitly prior-informed. The workflow makes within-phase revision and new-cycle replacement visible so both can be evaluated.

5.3 One stage can contribute to several components

Stage 4 produces assignment and outcome records for STAT-E, but it also produces delivery and setting information needed for SUB-E. Stage 3 develops parts of STAT-J through the randomization and analysis plan while developing parts of SUB-J through measurement and implementation choices. Stage 6 does not manufacture substantive justification after seeing the result; it checks how the planned support fits what happened, interprets each result, and

completes each bounded answer. Stage 7 then brings the completed question-level cycles together, gives every question an explicit outcome, and synthesizes across them without forming an omnibus argument.

This many-to-many relationship is why workflow cannot replace the argument components. The stages describe work; the components describe logical roles in the support for a particular answer.

5.4 What readers can and cannot infer from the documented workflow

A reader can usually trace the logical support more confidently than the historical workflow. A dated protocol can support the claim that a question, estimand, or analysis was prospective. Versioned code and deviation logs can show when changes occurred. A Methods section written after analysis cannot by itself establish timing.

A documented-workflow trace should therefore use restrained timing descriptions:

- **prospectively documented** when a dated record precedes the relevant evidence;
- **implemented as planned** when study records support correspondence between plan and conduct;
- **documented deviation** when the change and its timing are reported;
- **retrospectively described** when the paper gives the decision but not its timing; and
- **timing not located** when the accessible record does not establish when the decision occurred.

These are workflow-timing descriptions, not the four documentary statuses and not quality-evaluation judgments. They prevent a polished narrative from being mistaken for a chronological record. They also prevent an absent protocol from being treated as proof of poor research. The reader states what can be established from the materials; quality evaluation may then conclude **insufficient information to judge** for a particular quality question when the missing timing matters.

6 Extending Quality Evaluation for Coordinated Questions

6.1 Responsibility and use

As Chapter 1 explains, authors have the first responsibility to evaluate and document their arguments, while collaborators, supervisors, reviewers, editors, and readers independently evaluate them. Researchers can use the questions while planning, conducting, and reporting a study; readers and reviewers usually work from the accessible paper and related materials.

A judgment under one quality-evaluation question may be **adequate for this purpose**, **needs revision**, **insufficient information to judge**, or **not applicable**. The output is a reasoned

profile, not a summed score. One decisive weakness can require a claim to be withdrawn even when other features are strong.

6.2 The six basic questions plus Cross-question coherence

Chapter 1 introduced six quality-evaluation questions. Coordinated questions require one addition. The seven questions can be remembered in four groups without renaming them:

- **Alignment:** Question alignment.
- **Support:** Evidence integrity, Justification adequacy, and Inferential adequacy.
- **Translation:** Transition fidelity and Claim calibration.
- **Synthesis:** Cross-question coherence—do the question-specific arguments remain coherent with one another, and is their study-level synthesis warranted?

When one argument contains statistical strands, **Inferential adequacy** asks whether the strands jointly support the intended statistical inference; this includes their analytic status, multiplicity, dependence, and uncertainty, because parallel strands need not be statistically independent. The same question also evaluates the reasoning that integrates their results within SUB-I; it is not restricted to statistical procedure. **Transition fidelity** asks whether SUB-I preserves each strand’s result and uncertainty; and **Claim calibration** asks whether SUB-C retains the conditions and limitations of every contributing STAT-C. Relations among separate question–answer pairs belong instead to **Cross-question coherence**.

Suppose a draft conclusion for the upgraded study states:

Special Practice produced an educationally important improvement, especially for less-prepared students, and greater attendance increased the benefit.

The conclusion combines one supported result with three unsupported extensions: importance, effect variation, and attendance causality or mechanism. The evaluation profile identifies where the problems begin and what they require the writer to change.

Table 7

Seven quality-evaluation questions applied to an overstrong synthesis.

Group and quality-evaluation question	Beginner question	Judgment and required response
Alignment – 1. Question alignment	Do question, estimand, design, method, and intended answer agree?	Needs revision. The primary assignment question is aligned; the attendance sentence changes from association to causality and must be revised

Group and quality-evaluation question	Beginner question	Judgment and required response
Support – 2. Evidence integrity	What constitutes evidence, how was it produced, and what do missingness, dependence, and processing imply?	Insufficient information to judge. Records are largely adequate, but the missingness process is not fully documented, so support for inference to the all-randomized estimand remains conditional on the missingness analysis
Support – 3. Justification adequacy	Do design and statistical conditions support the statistical move, and do measurement and context support the substantive move?	Needs revision. Randomization supports assignment, not attendance; assessment meaning and one-setting scope remain bounded
Support – 4. Inferential adequacy	Do statistical procedure, uncertainty, checks, sensitivity, multiplicity, and substantive reasoning support the inferences within both layers?	Needs revision. The primary interval supports a positive effect but does not establish an effect above two points; the compatible interaction and attendance patterns do not resolve their directions
Translation – 5. Transition fidelity	Are magnitude, uncertainty, status, and reservations retained across layers?	Needs revision. The draft loses uncertainty and turns secondary point estimates into definite substantive patterns
Translation – 6. Claim calibration	Are claims no stronger than evidence, design, uncertainty, alternatives, and scope permit?	Needs revision. Replace “educationally important” with uncertainty about whether the effect exceeds the benchmark; remove subgroup and attendance causal claims
Synthesis – 7. Cross-question coherence	Are distinct estimands and outcomes preserved, and is synthesis transparent?	Needs revision. Keep the positive primary answer beside two inconclusive answers rather than making all three positive

The repaired synthesis is:

Assignment to Special Practice improved mean delayed-assessment performance for these participants under the stated conditions. Whether the effect exceeded

two points or varied with baseline preparation remains uncertain. Among observed-outcome cases, the attendance association was inconclusive; the exploratory analysis does not identify whether attending more sessions caused better performance.

Keep four outputs distinct. Reading records each **documentary status** and the authors' **reported answer**. Each quality question produces a judgment such as adequate for this purpose, needs revision, insufficient information to judge, or not applicable. Evaluation then states the **supported bounded answer** and one conclusion for each question–answer pair. The study-level synthesis is evaluated separately under Cross-question coherence.

For the three correctly bounded pairs in the question map, the supported answers match what was reported, so each is **supported as stated**. The causal attendance clause in the overstrong synthesis creates a different implied pair; because the accessible design provides no defensible causal attendance answer, that implied pair is **not supported**. The synthesis itself **requires revision** under Cross-question coherence; it does not receive one of the four pair-level conclusions. **Not assessable from the accessible record** remains available when the record is insufficient to decide a pair-level conclusion.

No total score is needed. The evaluation changes the answer directly without turning documentary presence into a quality judgment.

6.3 Locate the originating problem, then trace its consequences

Do not count one weakness several times merely because it affects several parts of an argument. Locate the originating problem under the question whose primary object contains it, then trace its consequences. Here the attendance mechanism first fails Question alignment because it changes an associational estimand into a causal one, and it fails Justification adequacy because attendance was not randomized. The unsupported move then propagates to Transition fidelity, Claim calibration, and Cross-question coherence. Those later questions show what the final wording and synthesis must repair.

Cross-question coherence is normally not applicable to a genuinely one-question study. If a conclusion introduces an unmasked subgroup, mechanism, or prediction claim, however, the study now contains another implied question–answer pair that must be separated and evaluated. In a coordinated study, Cross-question coherence tracks priorities, shared evidence, multiplicity, tensions, and synthesis across those distinct pairs.

7 How Regression-Based Methods Develop and Instantiate the Approach

Here **instantiate** means *put selected parts of the approach into practice*. Regression can express a declared comparison, help estimate its estimand, quantify uncertainty, organize checking, and

support a bounded statistical claim. It does not define the research question or estimand and does not replace the full argument.

7.1 From a two-group comparison to adjustment

Let Y be delayed score and Z indicate Special Practice assignment. A simple two-group regression is:

$$E(Y \mid Z) = \beta_0 + \beta_1 Z.$$

Here β_0 represents the Usual Practice mean and β_1 represents the Special-minus-Usual mean difference for the analysed records. This is the same comparison that can be calculated directly from two group means. Regression provides a common language; it does not create a new question.

For the primary analysis, baseline preparation X and section indicators can improve precision and respect blocked assignment:

$$E(Y \mid Z, X, S) = \beta_0 + \beta_1 Z + \beta_2(X - 20) + \gamma_S.$$

The reported primary result is not simply whichever coefficient appears in the output. Regression standardization averages fitted assignment comparisons over the target group. In this additive linear model it reproduces the constant assignment coefficient; interactions or nonlinear models can make the averaged comparison differ, and neither is the crude 3.1-point observed-case difference.

7.2 An interaction represents a different question

The secondary model adds an assignment-by-baseline product:

$$E(Y \mid Z, X, S) = \beta_0 + \beta_1 Z + \beta_2(X - 20) + \beta_3 Z\{(X - 20)/10\} + \gamma_S.$$

Now β_3 represents how the assignment contrast changes per ten-point higher baseline score under the specified scale and functional form. It does not estimate the causal effect of baseline preparation. It also does not prove that effect variation is absent when its interval includes zero.

Interaction is scale- and specification-dependent. A pattern on a mean-score scale can differ from a pattern on a probability or odds scale. Interpretation should stay within the supported baseline range and retain uncertainty.

7.3 Attendance uses the same language with different support

A regression for Pair 3 includes attendance alongside assignment, baseline, and section to estimate a conditional attendance association among the 300 observed-outcome cases. Attendance occurs after assignment. Conditioning on it can introduce post-assignment selection bias and can also change the interpretation of the assignment coefficient (Elwert & Winship, 2014). Randomization supports the unconditioned or appropriately standardized assignment comparison; it does not automatically support either coefficient in an attendance-conditioned model.

The regression is therefore used only for an exploratory association argument. It does not estimate a causal effect of attendance. Claim mode comes from the question and design, not the appearance of regression.

Table 8

How regression instantiates distinct parts of three arguments.

Regression move	Question-level role	Result in the upgraded study	What remains outside regression
Two-group indicator	Expresses the assigned-format comparison	Observed-outcome difference of 3.1 points	Missing-outcome estimand and wider scope
Baseline and section adjustment	Improves precision and respects blocked assignment while estimating the average estimand through standardization	Primary effect 2.9, 95% interval [1.9, 4.0]	Measurement meaning, missingness plausibility, implementation, and transport
Assignment-by-baseline interaction	Expresses effect variation over the baseline scale	-0.8 per ten points, 95% interval [-2.7, 1.1]	Adequacy of the moderator measure, functional form, and substantive meaning
Post-assignment attendance association	Expresses an observed-case conditional association	0.3 per session, 95% interval [-0.4, 0.9]	Causal effect of attendance, missing outcomes, and unmeasured reasons for participation

7.4 Where regression enters the statistical layer

Regression implements different parts of the statistical reasoning, but it does not create the study's empirical evidence:

- **STAT-E is the input.** The outcome, assignment, baseline, section, attendance, missingness, and processing records must exist before the model is fitted. The analysis must keep exclusions and transformations visible rather than treating model-ready data as the original evidence.
- In **STAT-J**, the model and estimator help connect a declared estimand to the records through choices about functional form, adjustment, dependence, missingness, uncertainty, and checking when the associated warrant is stated or implicit but traceable. A fitted model does not show that those choices or their conditions are adequate.
- In **STAT-I**, fitting, checking, and sensitivity analysis produce estimates, intervals, diagnostics, and comparisons that must be reasoned about for the declared estimand.
- In **STAT-C**, the study states its statistical conclusion about magnitude, direction, uncertainty, and conditions; evaluation later judges its calibration.

The substantive layer still has to situate the evidence, augment and contextualize the statistical justification with reasons about measurement and design relevance, interpret magnitude, consider alternatives, and state the substantive answer with its bounds. Quality evaluation is a separate task.

Regression cannot supply by itself:

- population representation;
- measurement validity;
- causal identification;
- a plausible missingness condition;
- implementation fidelity;
- practical or theoretical importance;
- mechanism; or
- transport to another setting.

These are not afterthoughts called “limitations.” They help determine whether the coefficient answers the question.

7.5 Checking, sensitivity, and claim revision

Regression makes several kinds of checking possible, but their role must remain bounded.

A residual display can reveal tension with a linear mean pattern or constant spread. An influence analysis can show that an estimate depends heavily on a few observations. Comparing adjusted and unadjusted estimates can reveal how the chosen specification changes the estimate

or precision, or whether the fitted procedure addresses a quantity different from the declared estimand. None of these checks proves that the model is true or that unmeasured problems are absent.

For Pair 1, useful checks include whether the baseline relationship is reasonably represented over the observed range, whether section-specific contrasts point in compatible directions, and whether the standardized result depends strongly on the missingness procedure. A serious failure should enter STAT-I and change STAT-C.

For Pair 2, plotting estimated contrasts over supported baseline values is more informative than announcing one interaction coefficient. Sparse data near the scale ends and alternative nonlinear forms can make an apparent subgroup pattern unstable. Because the interval includes materially decreasing, negligible, and increasing patterns, the reported answer remains inconclusive; width alone does not determine that conclusion. Exploratory alternatives should not be used to search for a preferred threshold.

For Pair 3, no residual check can remove unmeasured reasons for attendance. Sensitivity can show how large an omitted association would need to be to change the estimate, but the result remains observational. Checking improves calibration; it does not upgrade design.

This is why model checking belongs within inference rather than in a detached technical appendix. If a check matters, its consequence should be visible in the statistical claim and carried through the substantive layer.

7.6 Continuity with later chapters

An intercept-only model represents a mean. A group indicator represents a mean difference. Continuous variables express slopes or curves. Additional variables define conditional or standardized comparisons, and product terms express variation in comparisons. Later chapters develop these methods in detail.

The central discipline does not change: begin with a question, declare the estimand, preserve the design, make uncertainty and checking visible, interpret through the substantive layer, and end with a bounded answer, which may be inconclusive.

8 Transferring the Approach to a Published Study

8.1 The study and its materials

Deslauriers and colleagues compared active and passive instruction in two large-enrolment introductory physics courses (Deslauriers et al., 2019). The open-access paper and supporting appendix provide a compact opportunity to transfer the approach from a designed hypothetical record to actual research reporting.

The intervention occurred during two consecutive class meetings. Students were randomly assigned to two groups. In the first meeting, one group received active instruction on static equilibrium while the other received a passive lecture. In the second meeting, the instructional conditions were reversed for the fluids topic. The instructional condition therefore crossed over within two fixed group–instructor–room combinations. The study used both random assignment to group and a crossover of instructional condition.

The groups received identical content and handouts. In the active condition, students attempted problems in small groups before receiving complete solutions. In the passive condition, the instructor presented the same material and solutions. Of 157 enrolled students, 149 participated; eight opted out or did not complete the study.

After each class, students completed:

- a 12-item test of learning (TOL); and
- four feeling-of-learning (FOL) items, analysed using both a designated primary item and a principal-component-weighted composite of all four.

The tests were written from learning objectives by an author who did not see the course materials, and instructors did not see the tests. These design details contribute to measurement and comparison justification.

8.2 Two focal question–answer pairs

The paper contains at least two focal pairs:

1. What effect did active rather than passive instruction have on measured learning?
2. What effect did active rather than passive instruction have on students' feeling of learning?

The adjusted standardized regression results were:

- active instruction was associated with a **0.46 standard-deviation higher** TOL score than passive instruction; and
- active instruction was associated with a **0.56 standard-deviation lower** FOL score than passive instruction.

For each outcome, the reported test used zero as the null target value for the adjusted active-minus-passive standardized contrast and had $p < .001$. These tests belong in STAT-I beside their respective 0.46- and -0.56-SD estimates. As Chapter 1's numerical-result safeguard emphasizes, a p -value alone supplies neither direction, magnitude, nor precision; an omnibus test might not even supply direction. Chapter 3 develops the calculation and fuller interpretation of tests, intervals, and inconclusive results. The point estimates changed little when student covariates were removed or student fixed effects were added. The paper also considered peer effects and provided study materials in its supporting appendix.

The two results do not form one answer. They answer different questions with different outcomes and measurement meanings. Their study-level synthesis is that active instruction improved performance on the immediate tests while students felt that they learned less.

8.3 Tracing the supporting record

Table 9

Documentary traceability for two arguments in a published randomized study.

Pair and component	Documentary trace	Documentary status	Component content located
TOL – STAT-E	Methods, assignment description, crossover table, participation information, 12-item tests	Stated in the paper; test material located in related materials	Group assignment, crossover, participation, and immediate test records; adjusted-model analytic N after missing pretests is not located in the compact result table
TOL – STAT-J	Design description, model specification, robustness descriptions, supplement	Stated in the paper; additional checks located in related materials	Randomized crossover warrant, adjustment strategy, and alternative specifications
TOL – STAT-I	Regression Table 3 and robustness descriptions	Stated in the paper	Standardized 0.46 active-minus-passive estimate, test of a zero adjusted difference with $p < .001$, and stability across selected specifications
TOL – STAT-C	Results text, Table 3, and Figure 2	Stated in the paper	Reported positive adjusted comparison for immediate test performance
TOL – SUB-E/J/I/C	Learning objectives, course-material-blinded test preparation, matched content, Discussion	Stated in the paper; located in related materials (supplement)	Immediate topic-specific performance interpretation, instructional context, and reported scope
FOL – STAT-E	Four survey items, response scale, assignment and crossover records	Stated in the paper; item detail located in related materials	Immediate self-reported perceptions after each class

Pair and component	Documentary trace	Documentary status	Component content located
FOL – STAT-J	Composite description, model specification, and designated-item analysis	Stated in the paper	Composite warrant, adjustment strategy, and alternative item analysis
FOL – STAT-I	Regression Table 3 and designated-item analysis	Stated in the paper	Standardized -0.56 active-minus-passive estimate, test of a zero adjusted difference with $p < .001$, and consistency with the designated item
FOL – STAT-C	Results text, Table 3, and Figure 2	Stated in the paper	Reported negative adjusted comparison for immediate feeling of learning
FOL – SUB-E/J/I/C	Survey wording, discussion of fluency and metacognition, interviews	Stated in the paper; interview detail located in related materials	Immediate perceived-learning interpretation and proposed mechanism context
Cross-question synthesis	Abstract, Results, Discussion, figures	Stated in the paper	Higher measured performance and lower perceived learning are reported as distinct outcomes
Transport and mechanism context	Discussion, interviews, semester intervention, supporting literature	Stated in the paper; further material located in related materials	Proposed cognitive-effort explanation and discussion of possible extension beyond the two class meetings

8.4 What is strongly supported

Random assignment, the instructional-condition crossover within fixed group–instructor–room combinations, matched content, and common timing provide substantial support for comparing active with passive instruction in the implemented classes. The different directions for TOL and FOL are not a contradiction because the outcomes answer different questions.

The crossover also creates obligations. Each student experienced two ordered class meetings on different topics, so readers should examine topic, period, instructional order, and possible

carryover. Groups, instructors, and rooms remained fixed rather than being separately crossed. The randomized group assignment, instructional-condition crossover, and relevant indicators reduce some concerns, but they do not make those design conditions disappear.

The paper's use of student background measures, instructor and topic indicators, alternative specifications, and peer-effect discussion adds statistical justification. Course-material-blinded test preparation by a coauthor and the supplied materials make parts of the measurement support inspectable.

8.5 Where the answer should stop

The direct study population consists of students in two Harvard introductory physics courses. The intervention lasted two class meetings, and the outcomes were measured immediately. The TOL therefore supports a bounded claim about short-term performance on topic-specific tests, not durable learning across disciplines. FOL is a self-reported perception measure, not learning itself.

Under the crossover's topic, period, order, and carryover conditions, the randomized comparisons support effects of assigned instructional mode on TOL and FOL. They do not by themselves establish that increased cognitive effort caused the difference, that novice metacognition is the unique mechanism, or that the same effects will occur in every course. The interviews, literature, and later semester intervention provide relevant evidence, but those mechanism and transport claims require their own arguments.

A calibrated synthesis is:

In two introductory physics courses, active rather than passive instruction increased immediate topic-specific test performance while reducing students' immediate reported feeling of learning under the implemented crossover conditions. The study does not by itself establish long-term learning, a unique psychological mechanism, or effects in other populations and disciplines.

This transfer example illustrates why the accessible record must be read by logical function rather than paper section. It also shows why one paper can contain several related but distinct arguments. For the bounded TOL and FOL answers stated above, the supported answers match the reported answers, so the pair-level conclusions are **supported as stated**. Stronger claims about a unique mechanism, durable learning, or all disciplines are **not supported** as answers to those additional implied questions by the two focal randomized comparisons alone.

9 Summary and Reading Bridge

9.1 What the advanced unit adds

A relatively complex study may coordinate several questions without collapsing them into one argument. Each question–answer pair keeps its own estimand, support, analytic status, reported answer with its bounds, and question-level construction cycle. When several estimands are jointly necessary for one integrated answer, they can instead form statistical strands inside that one argument. The seven-stage extension adds prospective cross-question alignment to Stage 3 and a new Stage 7 that brings completed cycles together for explicit question outcomes and study-level synthesis. The seventh quality-evaluation question checks the coherence of that synthesis. Regression can instantiate selected comparisons and inferential moves, but the complete reading must still trace design, measurement, context, uncertainty, and scope.

9.2 Further extension for latent-variable questions

Observable scores already require measurement reasoning. When a question contains a latent variable or construct, the statistical part of its one question-level argument may be elaborated into linked measurement and structural layers rather than retained as one undifferentiated statistical layer.

A **measurement layer** addresses what observed indicators support about a latent variable, score, or classification. A **structural layer** addresses relationships among latent variables or between latent and observed variables. Each layer can contain Evidence, Justification, Inference, and bounded Claim.

Both layers elaborate the statistical layer and then connect to the substantive layer. This remains one argument for one question–answer pair, not separate measurement and structural arguments. Parallel component labels do not imply that every cross-layer operation is identical. The full architecture will be developed in the book’s latent-variable part.

9.3 Intellectual and practical reading bridge

The approach is a tutorial synthesis:

- Toulmin emphasizes evidence, warrants, qualifications, and reservations (Toulmin, 2003).
- LeMire offers a narrower statistical argumentation precedent (LeMire, 2010).
- Kane and the testing standards organize support for score interpretations and uses (American Educational Research Association et al., 2014; Kane, 2013).
- Estimand-first reasoning connects theoretical questions to estimands, identification, and evidence (Lundberg et al., 2021).
- PPDAC and GAISE emphasize iterative statistical enquiry, context, real data, and conceptual understanding (American Statistical Association, 2016; Wild & Pfannkuch, 1999).
- Design principles explain why representation, assignment, measurement, implementation, and transport bound claims (Shadish et al., 2002).
- Preregistration and reporting standards can strengthen prospective transparency and traceability without guaranteeing validity (Appelbaum et al., 2018; Nosek et al., 2018).

The approach is not offered as a complete new theory of inference. Its contribution is pedagogical integration: one argument for every question–answer pair, linked layers, a two-phase construction workflow with within-phase iteration, and quality-evaluation questions that lead directly to calibrated answers.

Exercises

Exercise 1: Separate the questions (Foundational)

A study asks whether a mentoring program:

1. changes average end-of-year score;
2. works differently for students with different baseline scores;
3. improves attendance; and
4. predicts which students will leave the course.

For each question, state:

- its intended claim mode;
- a plausible estimand;
- whether random assignment to mentoring directly supports the answer; and
- why it needs a distinct argument.

Exercise 2: Identify shared and question-specific support (Foundational)

For the upgraded special-practice study, classify the following as shared across all three arguments, shared across only Pairs 1 and 2, or primarily question-specific:

- participant consent records;
- within-section randomization;
- baseline score;
- session attendance;
- delayed outcome;
- missing-outcome condition;
- assignment-by-baseline interaction;
- attendance-confounding justification; and
- two-point importance benchmark.

Explain why “shared evidence” does not mean “identical adequacy.”

Then consider the alternative integrated decision question, “Did assignment meet the prespecified success criterion of a positive delayed-performance effect and no completion-rate reduction greater than five percentage points?” Explain why its score and completion estimands can be two statistical strands of one argument, and what would have to change for them to become two separately owed question–answer pairs.

Exercise 3: Diagnose an estimand–analysis mismatch (Foundational)

An analyst reports the observed-case difference of 3.1 points as the average effect for all 320 randomized students. Explain:

1. how the observed-case analysis and available evidence fail to match the declared all-randomized estimand;
2. which component or components are affected;
3. what condition or method is needed to connect 300 observed outcomes to the 320-person estimand; and
4. how the final claim should change if that connection is not defensible.

Exercise 4: Map the construction workflow (Integrative)

Use the seven-stage workflow: Stages 1–3 form Phase 1, and Stages 4–7 form Phase 2. Assign each event to one or more workflow stages:

- reviewing evidence about delayed assessment;
- declaring the two-point benchmark;
- fixing the 320-participant evidence-production plan and stopping rule;
- discovering a scheduling disruption;
- discovering during Phase 2 that the delayed assessment cannot answer the declared primary question;
- formulating the exploratory attendance question after evidence production had begun;
- documenting session attendance;
- revising a missingness sensitivity analysis;
- interpreting the interaction interval;
- integrating the three answer outcomes.

Identify (a) one justified return within the current phase, (b) one Phase 2 discovery that requires closing the current cycle and beginning a new or revised argument cycle, and (c) one new question whose timing requires a separate exploratory cycle.

Exercise 5: Apply the seven quality-evaluation questions (Integrative)

Evaluate this conclusion:

Special Practice is especially effective for less-prepared students and works because it makes students attend more sessions.

Give a judgment for all seven quality-evaluation questions. Use **adequate for this purpose**, **needs revision**, **insufficient information to judge**, or **not applicable**. Then write a replacement synthesis that preserves all three reported pair outcomes, including the inconclusive observed-case attendance association. State a supported bounded answer and pair-level evaluation conclusion for each original pair. Finally, identify the implied causal-attendance pair, give its pair-level conclusion, and say whether the study synthesis is coherent as written or requires revision.

Exercise 6: Assignment versus attendance (Integrative)

Draw two causal stories in words:

1. assignment affects attendance, which affects score; and
2. motivation affects both attendance and score.

Explain why an association between attendance and score can arise in both stories and why random assignment to the practice format does not randomize attendance. Propose a future design that would answer a causal attendance question more directly.

Exercise 7: Decode three regression estimands (Integrative)

For each model below, identify the question, estimand, key justification, and strongest possible claim:

1. delayed score regressed only on assigned format;
2. delayed score regressed on assigned format, baseline score, and section; and
3. delayed score regressed on assigned format, baseline score, section, and an assignment-by-baseline interaction.

State one condition under which the assigned-format coefficient would not equal the marginal average assignment effect without standardization.

Exercise 8: Published-paper traceability (Integrative)

Using the Deslauriers et al. paper and supporting appendix:

1. locate the two focal questions;
2. map one STAT component and one SUB component for each outcome;
3. identify where the randomized crossover design is documented;
4. compare the paper's direct numerical support for TOL and FOL;
5. identify one mechanism claim that needs a separate argument; and
6. rewrite one broad generalization as a bounded answer.

For each located item, use **stated in the paper**, **located in related materials**, or **implicit but traceable**; use **not located** when appropriate and flag contradictions separately. Record documentary status separately from quality: an item can be stated in the paper yet inadequately justified.

Exercise 9: Independent multi-question audit (Integrative)

Choose a quantitative paper with at least two coordinated research questions whose estimands, evidence, models, priorities, or interpretations are connected, and at least one related material such as a protocol, supplement, code archive, or data dictionary. Produce:

1. a question-to-argument map;
2. one complete eight-component map;
3. a shared-versus-question-specific support analysis;
4. a documented-workflow trace, showing within-phase returns and marking the start of a new cycle whenever Phase 2 work led to revision of a Phase 1 decision;
5. a seven-question quality-evaluation profile;
6. a study-level synthesis; and
7. a list of component content that is **not located**, followed separately by any whole-pair conclusion of **not assessable from the accessible record**.

Do not infer that a decision was prospective solely because it appears in the Methods section.

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